

EN 611: Durability Behavior of Energy Materials

Course Code	EN 611
Course Name	Durability Behavior of Energy Materials
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	UG/M.Tech./ MS/ PhD
Distribution	Compulsory for M.Tech. in Energy Engineering with specialization in Materials, and Elective for others
Approval	10th Senate (Jun 2016)
Recoded to	MT 304 by the 64th BoA (May 2026)
Discontinued	64th BoA (May 2026)

Course Contents

- **Static loading and Deformation behavior**

Distinctive load – elongation behavior of different types of materials under uni-axial loading, definition of stress and strain, stress-strain diagram, elastic behavior, yielding and plastic behavior of ductile materials; necking instability, Important design criteria based on stress and strain at ambient temperature; strain rate and its influence of stress-strain diagram at elevated temperature. [4 Lectures]

- **Durability under static loading**

Defects in materials; linear defects or dislocations – types, burgers vector, slip, slip planes and slip directions, cross slip and climb, movement of dislocations leading to plastic deformation, stacking fault and partial dislocations, strain hardening, grain boundaries, strengthening mechanisms; recovery, recrystallization and grain growth during heating of deformed materials. [6 Lectures]

- **Durability under Creep Deformation**

Deformation under static loading at elevated temperature - creep curve, mechanisms of creep, temperature dependence of creep, deformation mechanism maps, cavitations, stress rupture versus creep, extrapolation schemes; Development of materials for ultra supercritical boilers for application in furnace panels, superheaters, thick section components and steam lines. [8 Lectures]

- **Durability under Cyclic Loading**

Deformation under cyclic loading, high cycle fatigue – S-N curve, effect of mean stress, Miner rule, cyclic stress-strain curve, low cycle fatigue, strain life equation, effect of stress concentration and size of component, Design for fatigue, effect of temperature on fatigue; development of turbine materials. [8 Lectures]

- **Durability under corrosion**

Overview of corrosion, corrosion in water and steam, High temperature corrosion; corrosion under boiler flue gases, hot corrosion in gas turbine components, materials protection by coating. [6 Lectures]

- **Durability under Erosion and Cavitation**

Overview of wear, erosion and cavitation wear, thermally sprayed coatings and weldings for repair of hydro-turbines and pumps. [6 Lectures]

Text Books

1. George E. Dieter, **Mechanical Metallurgy**, McGraw Hill Book Company 1986
2. J. Lecomte-beckers, M. Carton, F. Schubert and P.J. Ennis (Editors), **Materials for Advanced Power Engineering**, Vol.53, Part-1, 2006.
3. Zaki Ahmed, **Principles of Corrosion Engineering and Corrosion Control**, Elsevier Science & Technology Books, 2006.
4. Peter J. Blau, **Friction and Wear transitions in Materials**, Noyes Publications, 1989